Copenhagen/Amsterdam, October 24-25, 2023, a group of 13 like-minded countries with significant expertise and programmes in quantum information science and technology (QIST) attended the fourth roundtable meeting on how to work together to advance the field and grow the global QIST ecosystem.



Denmark, led by the Ministry of Foreign Affairs, hosted the meeting with delegates from Australia, Canada, Denmark, Finland, France, Germany, Japan, Korea, the Netherlands, Sweden, Switzerland, the United Kingdom and the United States. The roundtable meeting in Copenhagen was followed by a workshop on “Quantum Technologies: promote and protect” in Amsterdam co-hosted by the Netherlands and the UK on October 26-27. The full programme had been planned in close collaboration between Denmark and the Netherlands.

In Copenhagen, the group discussed concrete ways to advance positive use-cases of QIST technologies and to strengthen research and innovation cooperation through existing multilateral funding calls like Eureka. Finally, the group also discussed a set of draft Guiding Principles that should guide the collaborative efforts to support a global quantum ecosystem informed by science (see at the end of this readout). QIST is evolving rapidly with great potential for breakthroughs in a wide range of areas – from the green transition and energy, health, and communications to transportation as well as defence and security. International collaboration is key to unleash the enormous potential of QIST as it will combine the required expertise, ingenuity, and creativity in order to expand humanity’s fundamental understanding of QIST. The dialogue on shared principles underscored the countries’ shared ambition of developing QIST underpinned by shared values such as freedom of inquiry, openness, transparency, honesty, equity, reciprocity, fair competition, objectivity, accountability, protection and enforcement of intellectual property, and democratic ideals.

As a part of the meeting, the group met with central actors in the Danish QIST ecosystem, touring some of the most important sites from the original Niels Bohr Institute to the newly established Deep Tech Lab - Quantum under the BioInnovation Institute.

In Amsterdam, the group discussed how to strike the right balance between ‘promote’ and ‘protect’. Many national governments have marked QIST as a critical technology that is relevant for both national and economic security. As a result, we have seen the emergence of national guidelines in the areas such as investment screening, international research collaboration, and protective security (as well as multilateral guidance such as the Five Eyes Five Principles of Secure Innovation). As these evolve, this group explored possibilities to lower the barriers to international collaboration: because we need the best people from different backgrounds to work on the technological challenges that still lie ahead. The group decided to use future meetings of this group for exchanging best practices about regulatory quantum-specific initiatives on investment screening and protective security – and their implementation. After the workshop, the group was updated on developments in other parts of the world and the quantum initiatives the World Economic Forum will be launching during the annual meeting in Davos. Also, the group visited the quantum sensing labs in Amsterdam and the startup and academic ecosystem at the House of Quantum in Delft.

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PRINCIPLES FOR A GLOBAL QUANTUM ECOSYSTEM INFORMED BY SCIENCE, AS DISCUSSED IN COPENHAGEN

support quicker scientific

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maintaining leadership for our

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2. Incorporate subject-matter expertise in policy discussions and decisions.

The general unfamiliarity with QIST among the public and policymakers lends itself to hype and misconceptions. The group can leverage QIST subject-matter expertise for policy discussions to facilitate decisions based on the best available science.

3. Share best practices and coordinate outreach in joint effort to grow the QIST talent base.

The demand for QIST talent is currently outpacing supply. Sharing best practices for increasing awareness of and training in QIST will enable more people to have the opportunity to participate. Coordinated outreach efforts are mutually beneficial and expand the reach of any one nation.

4. Promote research security, align and reinforce technology protection measures, and support a fair marketplace, to create a vibrant and trusted global QIST industry.

The QIST industry is globally interconnected. This ecosystem can be enhanced with a trusted supply chain and developing international standards that enable the adoption of new quantum technologies. It can be safeguarded by promoting research security and through multilateral protection measures informed by an understanding of the technology and its progress.

5. Plan for the deployment of quantum-resistant cyber infrastructure, such as quantum-resistant cryptography, to responsibly address the risks of QIST.

Any system that uses existing public-key cryptography standards could be vulnerable to an attack by a future large-scale quantum computer. To mitigate this risk, the group should alert decision makers to plan for transitioning cyber infrastructure to quantum-resistant cryptography.

6. Increase quantum awareness and readiness in the development and use of quantum-based technologies and applications.

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consistent with shared values and that benefit society.

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